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SURVEY

Advances in Biomedical Missing Data Imputation: A Survey

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ABSTRACT Ensuring data quality in biomedical sciences is crucial for reliable research outcomes, particularly as precision medicine continues to gain prominence. Missing values compromise data quality and can make it difficult to perform data-based studies. The origins of missing values in biomedical datasets are diverse, including experimental errors, equipment malfunctions, and variations in data collection protocols tailored to individual patient conditions. To address the complex nature of missing values and the unique characteristics of biomedical data, a diverse spectrum of computational imputation techniques has emerged. These methods range from traditional statistical analysis to more modern approaches such as discriminative machine learning models and deep generative networks. This survey paper provides a comprehensive overview of the extensive literature on missing data imputation techniques, with a specific focus on applications in genomics, single-cell RNA sequencing, health records, and medical imaging. We outline the fundamental principles underlying each imputation technique and present a detailed analysis of their advantages and disadvantages, categorized by missing data patterns. To aid practitioners in method selection, we offer practical recommendations based on critical factors such as dataset size, data type, and missingness rate. By synthesizing insights from existing literature, we provide a holistic perspective on the effectiveness of various imputation methods under different biomedical contexts, thereby facilitating informed decision-making for researchers and practitioners in applying imputation techniques to biomedical data processing.

INDEX TERMS Biomedical data imputation, data-centric AI, deep learning, genomics, health record, imputation, machine learning, medical imaging, missing value, single-cell RNA sequence.

I. INTRODUCTION

Missing data is a prevalent issue in many datasets across various domains. Data fields may be absent for a variety of reasons, including the fact that they were never collected or were lost during the acquisition, storage, or processing stages. This can significantly degrade the performance of statistical and machine learning models, and the omission of records or features with absent data can result in the loss of crucial information, thereby introducing bias into analyses and statistical studies. In biomedical science, where

individualized health and biological discovery are central focuses, the presence of missing values can become an important damaging factor. Information gaps originating from missing values impose considerable limitations on the utility of datasets and can make the application of statistical techniques and computational tools more challenging if they lack robustness to missing values. Consequently, addressing the task of missing data imputation has become the focus of numerous research studies, both in general tabular datasets and more specifically in the context of biomedical science.

Missing data patterns can be classified into three distinct categories: (1) missing completely at random (MCAR), where the occurrence of missing data is unrelated to both

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TABLE 1. Overview of general imputation techniques.

Technique	Method	Data type	Missing pattern	Required data	Imputation approach	Iterative	Feature-wise
ML	MICE	Mixed	MAR	Partially-observed	Multiple	Yes	Yes
	Missforest	Mixed	MAR	Partially-observed	Multiple	Yes	Yes
	MIXGB	Mixed	MCAR/MAR	Partially-observed	Multiple	No	Yes
	KNN	Mixed	MAR	Fully-observed	Single	No	No
	Sinkhorn	Mixed	MAR	Partially-observed	Single	Yes	No
	MIRACLE	Mixed	MCAR/MNAR/MAR	Partially-observed	Single	Yes	No
	ICE	Mixed	MAR	Partially-observed	Single	Yes	Yes
	HyperImpute	Mixed	MAR	Partially-observed	Multiple	Yes	Yes
GAN	GAIN	Mixed	MCAR	Partially-observed	Multiple	No	No
	MisGAN	Mixed	MCAR	Partially-observed	Single	No	No
	GAMIN	Mixed	MCAR	Partially-observed	Multiple	No	No
	PC-GAIN	Mixed	MCAR	Partially-observed	Single	No	No
	MA	Mixed	MCAR/MNAR/MAR	Partially-observed	Single	No	No
	MI-GAN	Continuous	MCAR/MAR	Partially-observed	Multiple	No	No
	SGAIN	Mixed	MCAR	Partially-observed	Single	No	No
	GAGIN	Continuous	MCAR	Partially-observed	Single	No	No
	SCIS	Continuous	MCAR	Partially-observed	Single	No	No
AE	MCFLOW	Mixed	MCAR	Partially-observed		No	No
	MIDA	Mixed	MCAR/MNAR	Fully-observed	Multiple	No	No
	HI-VAE	Mixed	MCAR	Partially-observed	Single	No	Yes
	MIWAE	Mixed	MAR	Partially-observed	Single/Multiple	No	No
	NOT-MIWAE	Mixed	MNAR	Partially-observed	Single/Multiple	No	No
	VAEAC	Mixed	MCAR	Partially-observed	Multiple	No	No
	AmNet	Mixed	MCAR/MNAR/MAR	Partially-observed	Single	No	No
	DAEMA	Mixed	MCAR/MNAR	Partially-observed	Single	No	No
Data-centric AI	DataFix	Mixed	MNAR	Fully-observed	Single	Yes	No

observed and unobserved features; (2) missing at random (MAR), where the probability of missing data depends on observed features but not on unobserved ones; and (3) missing not at random (MNAR), where missingness is influenced by both observed and unobserved features. For example, in a hospital setting, an equipment malfunction that results in the absence of vital sign recordings would constitute an MCAR scenario, as the missing data is unrelated to patient-specific factors such as health status or demographics. A case of MAR might involve patients who, due to severe pain – a recorded factor – are less likely to attend follow-up appointments. MNAR scenarios often appear in health records where the nature of a patient's illness directly influences the likelihood of certain data being recorded.

Zero, median, mean, mode, and deck imputation are among the most straightforward statistical methods for handling missing data [1]. These techniques, which fill the missing value of a given feature (i.e., data field) with a constant value (e.g., a “0” or the feature mean), are widely adopted due to their simplicity and computational efficiency. However, these approaches often produce unrealistic imputed values, which can distort the variance and correlation structure of the data, potentially leading to biased models if used for model training. To overcome these limitations, a range of more sophisticated approaches, including discriminative and generative imputation methods, have been developed to provide more accurate missing value imputation. Discriminative approaches model the conditional probability of each missing feature as a function of the non-missing data. State-of-the-art discriminative approaches include Multivariate Imputation by Chained Equations (MICE) [2], Missforest [3], and

Matrix Completion [4]. In contrast, generative approaches model the joint distribution of all features to impute missing data simultaneously. Early generative approaches include Expectation Maximization algorithms [5], while more recent approaches are based on deep learning, mainly Generative Adversarial Networks (GAN) [6], [7], Denoising AutoEncoders (DAE) [8], and Variational AutoEncoders (VAE) [9]. Additionally, some techniques leverage the foundation of generative models for data imputation, such as deep latent variable models (DLVMs) using the exact likelihood [10] or stochastic backpropagation and approximate inference [11]. Multivariate exploratory analysis provides an alternative method for estimating missing values. This approach focuses on obtaining maximum likelihood estimates of parameters despite missing values, using Principal Component Analysis (PCA) [12] for continuous data, Multiple Correspondence Analysis (MCA) [13] for categorical data, and Multiple Factor Analysis (MFA) [14] for mixed data. The inverse transformation of projected data is employed to fill in missing values, considering scenarios of MCAR or MAR.

Imputation methods are commonly classified into single imputation and multiple imputation methods. Single imputation methods generate a single estimated value for each missing data point, whereas multiple imputation methods propose several plausible values for each missing data point, resulting in multiple complete datasets derived from the original incomplete dataset. Multiple imputation techniques are generally preferred, as they account for the uncertainty inherent in the imputation process and enable more robust statistical inference. In these methods, each imputation is performed independently, and the results are then combined

TABLE 2. Strengths and limitations of general imputation techniques grouped by missing pattern.

Missing Pattern	Technique	Strengths	Limitations
MAR	MICE	- Models a conditional density for each feature. - Provides multiple imputations. - Captures feature relationships.	- Computationally intensive for large datasets. - Imputation quality depends on the chosen models for each feature.
	MissForest	- Captures non-linear feature relationships. - Robust to noisy features. - Provides error estimates for predictions.	- Changes in input will likely alter predictions. - Computationally intensive for large datasets. - Must be re-executed each time new data is imputed.
	KNN	- Considers the similarity between observations. - No model training required.	- Requires careful tuning of hyperparameters. - Computationally intensive for large datasets. - Ineffective in high-dimensional data. - Requires choice of k and distance metric.
	Sinkhorn HyperImpute MIWAE	- Leverages OT metrics to align distributions. - Automated model and hyperparameter selection. - No extra computational overhead from missing data. - Highly competitive for continuous data.	- Computationally intensive. - Computationally intensive. - Requires a large number of samples for highly accurate results.
MCAR/MAR	MIXGB	- Captures non-linear feature relationships. - Increases variance with predictive mean matching.	On small datasets, it can be outperformed in computation speed by MICE or MissForest.
	MI-GAN	Accurate MI in high-dimensional datasets.	Training instability.
MCAR	GAIN	- Models joint data distribution. - Can generate multiple imputations. - Captures complex patterns.	- Training instability. - Mode collapse issues. - May not handle high missing rates well.
	MisGAN	- Does not require fully-observed data during training. - Good performance at high missing rates.	- May fail to maintain correspondence between imputations and the ground truth. - Costly training of two GANs.
	GAMIN	- Generates multiple diverse imputations. - Good performance at high missing rates.	May fail for low structured or complex missingness.
	PC-GAIN SGAIN	- Exploits potential category information. - Simpler architecture than GAIN. - Fast convergence. - Training stability.	- Requires pre-training and clustering steps. - May not capture as much complexity as GAIN.
	GAGIN	- Captures fine-grained local details. - Effective for numerical and image data.	Complex architecture.
	SCIS	- Reduces training time of GAIN. - Efficient for large-scale incomplete datasets.	Performance depends on accurate sample size estimation.
	MCFLOW	- Preserves the semantic structure of data. - Effective at varying degrees of missingness.	Complex architecture.
	HI-VAE VAEAC	- Supports likelihood models for multiple data types. - Allows conditioning on any subset of features in the latent space.	- May fail to recover the marginal distributions. - Assumes that the encoding and decoding distributions are Gaussian.
	MIDA	Extracts overcomplete latent representations for improved imputation.	Requires careful tuning of hyperparameters.
	DAEMA	- Incorporates an input-oriented attention mechanism to focus on observed data. - Effective attention even with simple DAE.	Needs a sufficient amount of data to model the data distribution.
MNAR	NOT-MIWAE	Incorporates prior information about missingness patterns.	Sensitive to incorrectly learned missing model.
	DataFix	Minimizes distribution shifts.	Computationally intensive.
MCAR/MNAR/MAR	MIRACLE	- Preserves the causal structure during imputation. - Improves the imputation of baseline methods.	- May struggle with high-dimensional datasets. - Not suitable for non-causal domains. - Requires learning the causal graph.
	MA	- Augments data with minimal additional computational cost. - Can be integrated into any generative imputation model.	Depends on the quality of initial generator outputs.
	AimNet	Incorporates a modified dot-product attention mechanism to capture structural dependencies.	May fail on purely continuous datasets.

into a single aggregated outcome [15]. To perform multiple imputations, it is necessary to use methods that can introduce variation in subsequent imputations, producing slightly different imputation results at each iteration. Simple models such as linear or logistic regression, or deterministic methods based on matrix decomposition, do not account for variation among imputations.

Imputation methods can be categorized into iterative and non-iterative approaches, which differ in their treatment of

estimating missing values. Non-iterative methods involve a one-time estimation of missing values without iteration or repeated calculations [6], [8]. In contrast, iterative methods involve a sequential process where missing values are estimated multiple times until convergence is reached. Iterative methods typically use an initial estimate of missing values and update these estimates in each iteration based on the observed and imputed values [2], [16]. Additionally, imputation methods may vary in how they address missing

TABLE 3. Imputation technique recommendations based on dataset size, data type, and missing rate.

Technique	Dataset size	Data type	Missingness rate	Recommendations
Mean/Median, Mode	Any	Continuous (Mean/Median), Mixed (Mode)	Low	Consider alternative methods for complex datasets to preserve variability and correlations.
MICE, MissForest	Medium	Mixed	Moderate	Reduce computation speed by selecting a subset of features based on correlation with the target variable.
MIXGB	Large	Mixed	Moderate	Normalize features before distance calculations. Improve efficiency by focusing on key features or weighting neighbors.
KNN	Medium	Mixed	Moderate	
MIRACLE	Small	Mixed	Any	-
Sinkhorn	Medium	Mixed	Moderate	-
HyperImpute	Medium	Mixed	Moderate	-
GAIN	Large	Mixed	Moderate	Carefully tune hyperparameters to prevent convergence issues.
MisGAN	Large	Mixed	High	-
GAMIN	Large	Mixed	High	-
PC-GAIN	Large	Mixed	High	-
MA	Medium	Mixed	Any	-
MI-GAN	Large	Continuous	High	Ensure sufficient sample size to mitigate training instability.
SGAIN	Medium	Mixed	Moderate	Use SGAIN when computational resources are limited and faster convergence is needed.
GAGIN	Large	Mixed	High	-
SCIS	Large	Mixed	Moderate	-
MCFlow	Large	Mixed	High	-
MIDA	Large	Continuous	Moderate	Run the model multiple times to obtain multiple imputations.
HI-VAE, MIWAE, NOT-MIWAE	Large	Mixed	Moderate	Select the appropriate model based on the missing data mechanism: use HI-VAE for MCAR, MIWAE for MAR, and NOT-MIWAE for MNAR data.
VAEAC	Medium	Mixed	Moderate	-
AimNet, DAEMA	Large	Mixed	Moderate	-
DataFix	Large	Mixed	High	-
Beagle, fastPHASE, MaCH	Large	Genomic Sequences	Moderate	Use with high-quality reference panels and phased data for genotype imputation.
MMM	Large	Genomic Sequences	Moderate	Use when pedigree information is available, especially for traits with low heritability.
RF	Medium	Genomic Sequences	Moderate	Train models on a per-chromosome basis to improve performance and reduce computational demands.
AutoComplete	Large	Phenotypic	Moderate	Use for phenotype imputation in large biobank datasets.
MAGIC, KNN, ScImpute	Medium	scRNA-seq	Moderate	Suitable for reducing dropout noise but may lead to over-smoothing.
SAVER	Medium	scRNA-seq	Moderate	Recommended to prevent oversmoothing.
scIGAN	Small	scRNA-seq	High	Particularly effective for rare cell populations and small datasets with few genes exhibiting low expression or cell-to-cell variance.
Adaptive Filters	Large	Imaging Data	Medium	-
CNN	Large	Imaging Data	Medium	Apply for reducing noise while preserving structural details. Ideal for tasks requiring detailed image reconstruction and restoration.
I2-GAN	Medium	Imaging Data	Medium	Use key features of slices near missing information as conditional variables to infer missing slices.
TTSR	Medium	Imaging Data	Medium	Employ for super-resolution tasks that restore texture and fine structural details.

values: some techniques handle imputation feature-wise [2], [17], addressing one feature at a time, while others handle all features simultaneously [18], [19].

This is a survey article aimed to summarize and highlight empirical evidence from existing studies to support recommendations on missing data imputation techniques. In this paper, we provide a comprehensive overview of these techniques, encompassing both widely adopted methods and recent innovations. We begin by discussing general imputation techniques (Section II), which include advances in machine learning (Section II-A), generative adversarial networks (Section II-B), autoencoders (Section II-C), and emerging data-centric AI methods (Section II-D).

We then focus on specialized imputation techniques in biomedical domains (Section III), covering areas such as genomic imputation (Section III-A), phenotype imputation (Section III-B), single-cell RNA data imputation (Section III-C), health record imputation (Section III-D), and medical image imputation (Section III-E). Following an in-depth examination of these approaches and a thorough analysis of their strengths and limitations, we provide a summary of the proposed imputation techniques and offer informed recommendations on their suitability and effectiveness, taking into account factors such as data types, dimensions, missingness patterns, and missing rates.

TABLE 4. Imputation and visualization libraries in Python/R.

Name	Implemented techniques	Language	Type	URL
sklearn.impute	Mean, Median, Mode, MICE, Missforest, KNN, Discriminative-based methods	Python	Imputation	https://scikit-learn.org/stable/modules/impute.html
fancyimpute	Factorization-based methods, KNN, MICE	Python	Imputation	https://pypi.org/project/fancyimpute/
HyperImpute	Mean, Median, Mode, Missforest, ICE, MICE, SoftImpute, MIWAE, MIRACLE, GAIN, Sinkhorn, EM	Python	Imputation	https://github.com/vanderschaarlab/hyperimpute
missingpy	KNN, Missforest	Python	Imputation	https://github.com/epsilon-machine/missingpy
impyute	KNN, MICE, EM, Mean, Mode, Median, Random	Python	Imputation	https://github.com/eltonlaw/impyute
statsmodels.imputation	MICE, MI, Bayesian Imputation with Gaussian Model	Python	Imputation	https://www.statsmodels.org/stable/imputation.html
MIDASpy	MIDAS	Python	Imputation	https://github.com/MIDASverse/MIDASpy
AutoImpute	Mean, Median, Mode, Random, Norm, Linear Regression, Binomial Logistic Regression, Multinomial Logistic Regression, Logistic Regression, Stochastic regression, Bayesian Linear Regression, Bayesian Binary Logistic Regression, Predictive Mean Matching, Local Residual Draws, MICE	Python	Imputation/Visualization	https://pypi.org/project/autoimpute/
missingno	Matrix, Bar, Heatmap, Dendrogram	Python	Visualization	https://github.com/ResidentMario/missingno
scikit-na	Heatmap, Stairs Plot, Histogram	Python	Visualization	https://github.com/maximtrp/scikit-na
PyMC	Maximum Likelihood Estimation and Bayesian Optimization Techniques	Python/R	Imputation	https://www.pymc.io/projects/examples/en/latest/case_studies/Missing_Data_Imputation.html
h2o	Mean, Median, Mode	Python/R	Imputation	https://docs.h2o.ai/h2o/latest-stable/h2o-docs/data-munging/imputing-data.html
gcimpute	Gaussian Copula Model, Low Rank Gaussian Copula Model	Python/R	Imputation	https://github.com/udelgroup/gcimputeR
SoftImpute	SoftImpute	R	Imputation	https://cran.r-project.org/web/packages/softImpute/index.html
MICE	MICE	R	Imputation	https://cran.r-project.org/web/packages/mice/index.html
Missforest	Missforest	R	Imputation	https://cran.r-project.org/web/packages/Missforest/index.html
missMDA	PCA, MCA, MFA	R	Imputation	https://rmisstastic.netlify.app/packages/missmda/
rMIDAS	MIDA	R	Imputation	https://github.com/MIDASverse/rMIDAS
missCompare	Mean, Median, Random, MI, MICE, Missforest, PCA, MCA, MFA	R	Imputation/Visualization	https://cran.r-project.org/web/packages/missCompare/index.html
VIM	Univariate, Bivariate, Multiple and Multivariate plot methods	R	Visualization	https://cran.r-project.org/web/packages/VIM/index.html
naniar	Missing value plots and examination of imputations	R	Visualization	https://cran.r-project.org/web/packages/naniar/index.html

Table 1 presents a detailed comparison of various imputation methods for general tabular datasets, categorized by key features such as data type, missingness pattern, the extent of observed data (fully or partially observed), the imputation approach (single or multiple), the involvement of iterative processes, and whether imputation is applied on a feature-wise basis. Table 2 further elaborates on these methods by summarizing their strengths and limitations, grouped according to the missing data patterns they are designed to handle. This table offers insights into trade-offs such as computational demands and the ability to model non-linear relationships. Table 3 further aids in method selection by offering recommendations based on key dataset characteristics such as size, data type, and the rate of

missingness. It also includes additional considerations and advice on when and how to apply these methods effectively, ensuring practitioners can choose the most suitable approach for their specific scenarios. Furthermore, Table 4 compiles a collection of Python and R resources dedicated to the management and exploration of missing data, including tools for both missing data imputation and visualization.

II. GENERAL IMPUTATION TECHNIQUES

A. MACHINE LEARNING

Discriminative approaches that use machine learning models to estimate missing values based on complete features have been widely used. These include parametric models

like Multivariate Imputation by Chained Equations (MICE) [2] and decision tree models like Missforest [3] and MIXGB [20]. While these approaches have demonstrated good performance across many applications, they can be computationally and memory intensive in some settings.

MICE [2] begins by imputing all missing values with simple placeholders – typically, the mean for continuous features and the most frequent value for categorical ones –. It then selects one feature as the target, reverts the placeholders for that feature to missing, and uses the remaining features to model the target's conditional density through a machine learning model. This iterative process continues over multiple cycles for each feature with missing values, progressively training models on better-imputed data and accounting for uncertainty by producing multiple imputations. Fig. 1 provides a general overview of how MICE operates. MICE is a variation of the Fully Conditional Specification (FCS) method [21], [22], which imputes missing values by defining separate conditional densities for each feature based on observed values from other features. FCS employs regression models in iterative cycles to predict missing values and has been effectively applied in methods like SRMI [23]. Another approach, joint modeling [21], uses a Bayesian framework to draw imputations from the posterior predictive distribution while considering feature relationships. Previous studies [24] have successfully used joint modeling for imputing missing values. One of the advantages of MICE is its ability to handle mixed data types by selecting appropriate models for each input. By modeling a different conditional density for each feature, MICE often outperforms other multivariate imputation models, such as those fitting a joint normal distribution for all features [24]. However, the computational complexity of performing multiple cycles can pose a challenge. This issue can be mitigated by selecting a subset of independent features based on their correlation with the target. MICE assumes that missing data is MAR; therefore, using it in other settings may result in skewed estimations and incorrect predictions. ICE [17] is an implementation of MICE for Stata that includes linear, logistic, multinomial logistic, and ordered logistic regression models.

Missforest [3] is a widely adopted version of MICE, wherein random forests are used to iteratively impute missing data. Random forests offer several key advantages over other models: they can handle both categorical and continuous features, model non-linear relationships within the data, and are robust to noisy features due to their built-in feature selection mechanisms. Additionally, they use bootstrap sampling, which provides an out-of-bag error estimate for their predictions. However, similar to MICE, the primary drawback of MissForest is the computational time required for imputation, which increases with the number of instances, the number of features, and the proportion of features containing missing values. Due to its iterative algorithmic nature, MissForest must be re-executed each

time new observations with missing data require imputation, which can be impractical in real-time applications.

MIXGB [20] is a recently developed imputation technique that employs extreme gradient boosting machines (XGBoost) as the base model to iteratively impute missing features. Although MIXGB is non-iterative by default, multiple iterations can be performed in a manner similar to the approach used in MICE. MIXGB incorporates subsampling and predictive mean matching to mitigate bias and enhance the accuracy of imputation results. While MIXGB may be slower than MICE or MissForest for small datasets, it is more scalable and computationally efficient for larger datasets.

K-Nearest Neighbors (KNN) [25] imputation is another popular method that fills in missing data based on the k -nearest neighbor observations in the training set. Given a distance metric between samples (e.g., Euclidian distance) and a pre-specified number of neighbors k , each sample with missing data is compared to a reference set of samples, and a weighted average of the k neighbors is used to predict the imputed values. An iterative approach similar to MICE can be combined with KNN to improve imputation accuracy. The selection of an appropriate distance measure and the number of neighbors is crucial for optimal performance [26]. Before computing distances, KNN typically requires data normalization and the conversion of categorical features to one-hot encoding or other embeddings. KNN is computationally intensive for large datasets due to the high complexity of calculating distances between all neighbors, and it may be ineffective in high-dimensional data as it struggles to identify proper neighbors. To reduce computational time or improve results, modifications such as focusing on a subset of features or weighting the neighbors can be applied [27], [28].

Matrix completion (MC) algorithms [4] aim to impute missing data by approximating the incomplete data matrix with a low-rank representation. Recently, there have been several approaches to this problem such as Polynomial Matrix Completion (PMC) [29], Non-Linear Matrix Completion (NLMC) [30], and Low-Rank Gaussian Copula (LRGC) [31]. These methods differ in their optimization problem formulations and the constraints they impose. They represent an evolution of previous Low-Rank Matrix Completion (LRMC) techniques [32], [33], [34] and have demonstrated effective imputation results when handling noisy data. SoftImpute [35] is another method specifically designed for matrix completion and low-rank singular value decomposition, which uses a soft-thresholding technique to estimate missing values.

MIRACLE [36] is a causally aware imputation algorithm designed to preserve the causal structure of the data when imputing missing values. It begins with an initial imputation from a baseline method and iteratively refines it by learning a missingness graph. Regularization techniques are applied to ensure that the imputation aligns with the underlying causal graph responsible for data generation. While effective, MIRACLE is unsuitable for non-causal domains such as

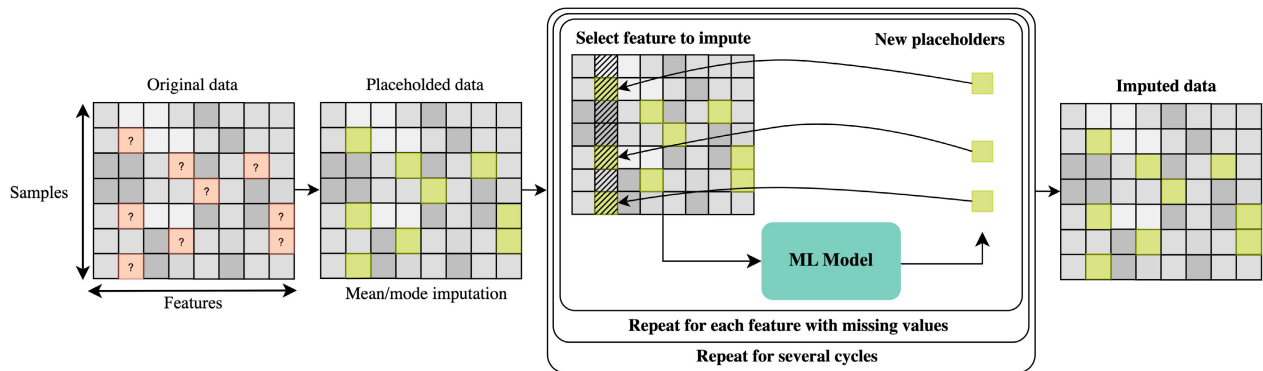


FIGURE 1. Overview of MICE.

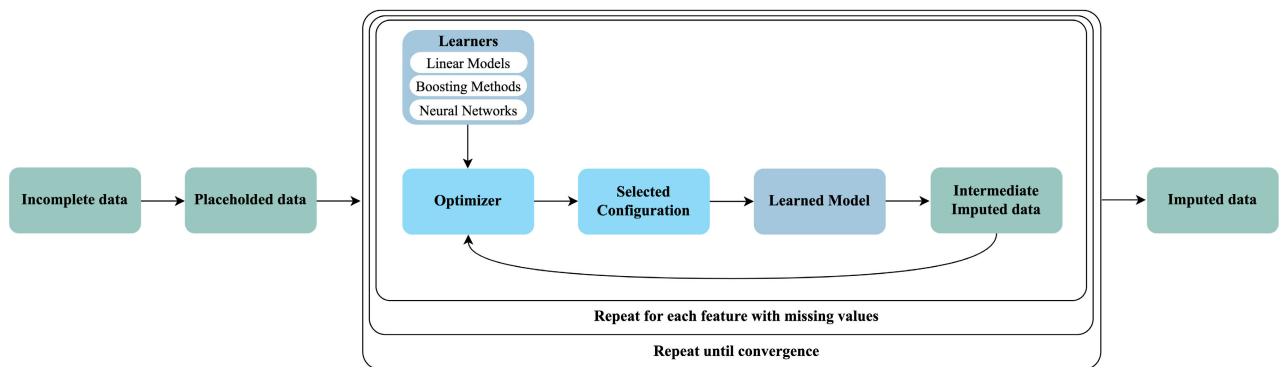


FIGURE 2. Overview of HyperImpute.

computer vision and may struggle with high-dimensional datasets due to its network architecture and causal discovery regularizer. Potential bias introduced by the baseline method is a concern; however, the algorithm generally performs well even with simple initial imputations like the mean.

The Sinkhorn imputation method [37] leverages optimal transport (OT) metrics, specifically Sinkhorn divergences, to measure the dissimilarity between batches of complete and incomplete samples. By aligning the distributions of these batches, the method generates imputations for missing data. Three variants of the Sinkhorn method have been developed: two parametric approaches, Linear Ridge Regression (Linear RR) and Multilayer Perceptron Ridge Regression (MLP RR), and one non-parametric approach (direct imputation method). The Linear RR variant, similar to MICE but using an OT loss function, and the MLP RR, which employs neural networks, demonstrate comparable or superior performance to MICE, highlighting the effectiveness of Sinkhorn loss in parametric models. Notably, the direct imputation method stands out as the best-performing OT-based approach.

In addition to existing imputation methods, HyperImpute [16] is an effective imputation framework that automates model and hyperparameter selection for each feature, removing the need for manual specification. Initially, it replaces missing values with the mean or median for continuous features and the most frequent value for categorical features.

It then employs an inside-out search strategy and an optimizer to identify the optimal learner configuration for each feature. This iterative process is repeated until convergence, as illustrated in Figure 2. HyperImpute is designed to handle incomplete data without assuming randomness in missingness patterns, offering a flexible and effective imputation framework.

B. GENERATIVE ADVERSARIAL NETWORKS

Generative adversarial networks have been broadly adapted in recent years for imputing missing data. One of the most popular methods is the Generative Adversarial Imputation Network (GAIN) [6], which generates plausible synthetic data through adversarial training. The architecture of GAIN, illustrated in Fig. 3, is composed of a generator and a discriminator network. The generator receives the incomplete data as input, a mask matrix indicating which components are missing, and a source of noise (typically Gaussian), and outputs the predicted imputed data. The discriminator takes the imputed data and a hint matrix as input and attempts to determine which components were originally observed and which were imputed by the generator. The generator is trained by minimizing a loss function that combines a reconstruction loss – computed using the predicted imputed data, the original data, and the hint matrix – and an adversarial loss from the discriminator network. The hint matrix, which reveals

partial information about the missingness of the original data to the discriminator, forces the generator to generate samples according to the true underlying data distribution. By modeling the distribution of the data rather than just the expectation, multiple imputations can be generated, capturing the uncertainty of the imputed values. GAIN is particularly effective for datasets with a small number of features and has proven to outperform methods like MICE, Missforest, and Matrix Completion. However, it may converge to zero or mean imputation for high missing rates [19], and its training process can be unstable [7].

Several models have evolved from GAIN to enhance performance by modifying its structure or loss function. MisGAN introduces data augmentation by using two generators to create synthetic complete data and missing masks, along with two discriminators: one models the missing data distribution, and the other makes fake missing data indistinguishable from real incomplete data. The generator of the imputation network fills in missing data to deceive its corresponding discriminator, which differentiates between imputed and synthetic complete data. While MisGAN outperforms GAIN at high missing rates, it requires training two generative adversarial networks, significantly increasing computational demands and complexity. Additionally, MisGAN sometimes fails to maintain correspondence between imputations and the ground truth [19]. The Generative Adversarial Multiple Imputation Network (GAMIN) [19] addresses this by generating multiple, diverse imputations, outperforming both GAIN and MisGAN for high missing rates. GAMIN, based on MisGAN, uses the missing data mask itself instead of a mask generator. It learns stochastic imputation candidates by initially replacing missing data with a prior imputation from an unconditional generator, which is then refined by a conditional generator into a more probable imputation. A novel confidence-based loss function measures the amount of adjustment required by the conditional generator to modify the prior imputation. During inference, the top- k imputations are selected based on their confidence scores.

PC-GAIN [38] is an extension of the GAIN model that enhances data imputation by incorporating potential category information. It involves pre-training the generator and discriminator on a subset of data with a low proportion of missing values, generating synthetic pseudo-labels through KMeans++ clustering [39], and integrating a classifier into the generative adversarial framework to infer pseudo-labels from the imputed dataset. This method enables the generator to produce more accurate imputation results than GAIN, particularly at high missing rates.

Missingness Augmentation (MA) [40] is a technique that enhances imputation results by dynamically augmenting data during the training of generative imputation models. MA achieves this by using the generator's outputs as new incomplete samples before each training mini-batch. This technique can be integrated into any generative adversarial

framework without modifying its structure, simply by applying a special reconstruction loss to the augmented samples. This loss, combined with the original loss function, constitutes the final optimization objective. MA has demonstrated superior performance compared to GAIN and MisGAN across various missing data mechanisms, with minimal additional computational cost. Furthermore, incorporating MA into MisGAN enhances its ability to recover the texture of missing regions in image data.

The Multiple Imputation via Generative Adversarial Network (MI-GAN) [41] is an ensemble of GANs that is used to impute missing values in high-dimensional, block-wise datasets under both MAR and MCAR mechanisms. In this method, each GAN within the ensemble is trained to impute a specific pattern of missing data. MI-GAN has demonstrated superior performance over other imputation methods in terms of statistical inference accuracy and computational speed. However, it faces limitations such as unstable training processes and the potential for non-convergence of generators, particularly when the training sample size is small.

To mitigate the challenges GAIN faces – such as unstable optimization, high computational costs, vanishing gradients, and mode collapse [42], [43] – Slim GAIN (SGAIN) and two Wasserstein variants, WSGAIN with weight clipping (WSGAIN-CP) and WSGAIN with gradient penalty (WSGAIN-GP), were developed [44]. SGAIN simplifies the original architecture by removing the hint matrix and adopting a slimmer design. It employs hyperbolic tangent activation functions in the output layers of both the generator and discriminator to achieve faster convergence. Unlike GAIN, SGAIN invokes the discriminator twice, once for real data and once for fake data. All three models – SGAIN, WSGAIN-CP, and WSGAIN-GP – achieve performance comparable to GAIN in terms of data imputation quality and post-imputation accuracy while being considerably faster, particularly SGAIN.

Although GAN-based methods can capture global information by approximating the distribution of generated data to that of real data, they often struggle to capture finer details such as the smoothness of homogeneous boundaries. To overcome this limitation, [45] proposed the Generative Adversarial Guider Imputation Network (GAGIN) for unsupervised imputation. GAGIN integrates a Global-Impute-Net (GIN) for capturing the global data distribution and performing initial imputation, a Local-Impute-Net (LIN) for refining local regions, and an Impute Guider Model (IGM) to guide the LIN based on the GIN results, ensuring focus on small, missing-centered areas. By effectively combining global and local information, GAGIN outperforms GAIN, MisGAN, and GAMIN across various missing rates on both numerical and image datasets.

GAN-based methods often incur high computational costs when calculating gradients using the mini-batch strategy for both the generator and discriminator in large-scale incomplete datasets. To alleviate this, [46] introduced the

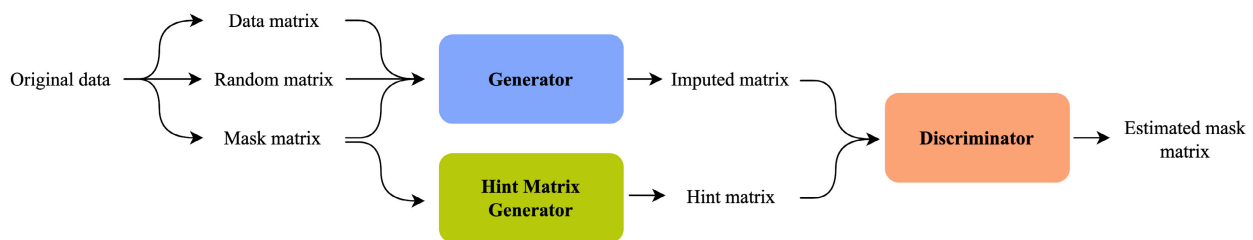


FIGURE 3. Overview of GAIN.

SCalable Imputation System (SCIS), which significantly reduces training time for these models. SCIS comprises two modules: the Differentiable Imputation Modeling (DIM) module and the Sample Size Estimation (SSE) module. The DIM module employs optimal transport theory to make GAIN differentiable, thereby addressing the vanishing gradient problem. The SSE module determines the optimal sample size for efficient and accurate training of the differentiable model. SCIS matches or surpasses the imputation accuracy of GAIN while requiring fewer training samples.

While GANs have been effective for data imputation, alternative generative models also demonstrate significant potential. For instance, MCFlow [47] combines normalizing flow generative models with Monte Carlo sampling to efficiently handle missing information in diverse datasets. It addresses the causality dilemma inherent in training with incomplete data by employing an iterative learning scheme that alternately updates the density estimate and the values of the missing entries. This approach makes MCFlow a powerful tool for modeling distributions from incomplete datasets with varying degrees of missing values.

Despite the advancements and successes of GAN-based imputation methods, they are not without limitations. Challenges such as handling mixed data types, tuning complexities, and convergence issues can hinder their performance. Additionally, GANs often rely on complete datasets for training, which can lead to biased results when the data inherently contain missing values. These limitations have prompted researchers to explore alternative approaches that can overcome these challenges while effectively imputing missing data.

C. AUTOENCODERS

In contrast to GANs, autoencoders (AEs) offer a different paradigm for data imputation by encoding input data into latent representations and reconstructing it using a decoder in an unsupervised manner. This mechanism allows autoencoders to extract essential information from the data even when missing values are present. To facilitate training on incomplete data, an initial imputation is usually performed, where missing continuous features are replaced with their mean values, and categorical features are filled with their most frequent values. AEs are well-suited for handling mixed

data types and can mitigate some of the convergence and tuning issues associated with GANs.

Denosing Autoencoders (DAEs) are a specialized form of autoencoder designed to reconstruct clean outputs from noisy inputs. Since missing data can be considered a form of noise, DAEs are well-suited for imputation tasks. Multiple Imputation with Denosing Autoencoders (MIDA) [8] extends this concept by using an overcomplete DAE. This model maps the input into a latent representation through successive hidden layers that are significantly larger in dimensionality than the input layer. The overcomplete architecture effectively addresses the limitations of conventional DAEs, which may struggle with imputing missing data due to their reliance on latent representations not directly observable in the input space. MIDA achieves multiple imputations by running the model multiple times, each initialized with different random weights.

The Heterogeneous-Incomplete VAE (HI-VAE) [9], along with MIWAE [48] and NOT-MIWAE [49], are advanced Variational Autoencoder (VAE) models designed to handle mixed data types and maximize the evidence lower bound for incomplete data under various missing data mechanisms – specifically, MCAR, MAR, and MNAR, respectively. HI-VAE enhances training stability and convergence by applying batch normalization to the input of the reconstruction model for numerical features and batch denormalization to the output layer of the generative model. By using a joint deep neural network to construct output distribution parameters, HI-VAE effectively captures the intricate relationships among features. It excels in imputing missing data through the trained generative model or inferred output distribution parameters, often outperforming methods like MICE and GAIN, especially in the imputation of categorical features.

The Variational Autoencoder with Arbitrary Conditioning (VAEAC) [50] is a notable extension of the VAE framework that allows conditioning on any chosen subset of features within the latent space, thereby enriching the imputation process with valuable context. This model is capable of handling both continuous and categorical data. It employs a neural network-based probabilistic model with Gaussian latent variables trained via stochastic variational Bayes. VAEAC demonstrates robust performance in both missing data

imputation and image inpainting tasks, often outperforming competitive methods such as MICE, MissForest, and GAIN.

AimNet [51] and DAEMA [52] are advanced imputation methods that incorporate attention mechanisms into AE architectures. AimNet employs a dot-product attention mechanism to create unified representations for imputing both continuous and discrete data. In contrast to AimNet, which applies the attention mechanism to individual features, and most imputation algorithms that learn from the full incomplete data including placeholder values, DAEMA uses an input-oriented attention mechanism in the encoder to focus on observed data. The attention mechanism in DAEMA is based on three components: a feature encoder that creates multiple estimation values for each latent feature, a feature selector that assigns greater attention to the most informative values, and a sum aggregation that combines feature vectors and selection vectors. Overall, DAEMA outperforms methods like DAE, AimNet, and MIDA in handling various missing data scenarios. AimNet tends to underperform compared to MissForest on purely continuous datasets.

An innovative approach involves using the average code from the latent representation of an AE to obtain a more robust latent representation [53]. This approach reconstructs missing data by decoding the average code and integrates multiple imputation within the latent space, unlike MIDA, which performs multiple imputation at the output layer. This method involves training a single AE, using smaller latent representations to reduce computational time, and employing random values as placeholders that change with each epoch for regularization purposes. This strategy addresses the issue of imputing constant values, which can lead to generalization problems as the model may memorize the initial imputed values. This method has demonstrated superior performance over MIDA and GAIN, particularly at high missing rates.

D. DATA-CENTRIC AI

Imputing missing values is central to data-centric AI, which focuses on improving both the quality and quantity of data to enhance AI performance. Missing values may arise due to the absence of information [54] or the complete unavailability of certain features when combining data from diverse sources [55]. Traditionally, imputation techniques have centered on minimizing imputation errors. A novel method called DataFix [18] employs adversarial learning principles to minimize the distribution shift between complete and incomplete samples. DataFix generates a range of replacement values for each incomplete sample from a limited search space derived from complete samples. Multiple discriminators, implemented as CatBoost classifiers [56], are used to distinguish between complete and imputed samples, and the replacement values that minimize the empirical likelihood of being an imputed sample are selected. This process repeats iteratively until no divergence

is detected between the distribution of complete samples and the distribution of imputed samples.

III. SPECIALIZED IMPUTATION TECHNIQUES IN BIOMEDICAL DOMAINS

A. GENOMIC IMPUTATION

In genomics, imputation is essential for correcting collection errors, inferring missing genetic markers, and integrating diverse datasets to ensure that all samples share genotype data for the same variants. This process relies on genotype information from a reference panel to estimate missing variants in a target panel, requiring high-quality reference genomes with linked haplotypes. Genomic imputation techniques must be both fast and memory-efficient to handle the vast scale of genomic sequences effectively. By assuming phased input data, the task simplifies to imputing missing alleles on a haplotype, which significantly reduces computational time [57]. The quality of imputation can be evaluated using several performance measures, including the intra-class correlation coefficient for heritability via average information restricted maximum likelihood, the correlation between observed and predicted values adjusted by the square root of heritability [58], and the difference between observed and expected heterozygosity proportions.

A straightforward method for imputing missing genomic data is to replace missing values with the mean allele frequency across reference samples at each locus. However, this approach reduces both global and sample-specific variability and disregards linkage disequilibrium among nearby markers, leading to a dataset that diverges from its true distribution. This deviation is often observable in Principal Component Analysis (PCA), where individuals with imputed values may shift from their actual population clusters toward the origin of the PCA plot, resembling admixed individuals. Such artifacts can result in misinterpretations of population structure, especially when missing data is frequent and non-random [59]. To address these limitations, specialized techniques have been developed to impute missing genomic data and correct sequencing errors while accounting for genetic linkage and marker dependencies. These methods primarily use Hidden Markov Models (HMMs) [60], [61] or linear models [62], [63], which exploit correlations among adjacent genetic markers and use information on marker order and phase. Unlike traditional imputation approaches such as k-Nearest Neighbors (KNN) [64], Single Value Decomposition (SVD) [65], or Random Forest (RF) [3], which may not fully capture the sequential and dependent nature of genomic data, these specialized methods are specifically designed to account for the inherent structure and dependencies in genomic sequences. In the case of RF, when applied to genomic data, it is recommended to train models on a per-chromosome basis to create more informative trees and reduce computational demands.

HMM-based approaches are among the most effective techniques for genotype imputation. These methods

predict the most likely path of haplotypes by modeling transition probabilities between ordered genetic markers. Popular HMM-based software tools include Beagle [61], fastPHASE [60], and MaCH [66], all of which use the Expectation-Maximization (Baum-Welch) algorithm for iterative parameter updates. Unlike MaCH, which models genotypes as discrete allele counts, Beagle and fastPHASE focus their HMM calculations on clusters of genotyped markers in the target data. The underlying concept is that individuals within a population tend to have haplotypes that cluster into groups of similar sequences within short chromosomal regions. While fastPHASE uses a pre-specified number of clusters, Beagle dynamically adjusts the cluster count to better reflect local patterns of linkage disequilibrium [67].

An alternative approach for genotype imputation involves leveraging the pedigree relationship matrix and marker covariance to incorporate genealogical information [68]. Pedigree-based imputation methods do not require a reference dataset that includes all populations under study. Multivariate Mixed Models (MMM) operate under the assumption that marker inheritance follows Mendelian patterns, thereby adhering to the known pedigree structure. MMM calculates the numerator relationship matrix based on the expectation of Mendelian allele inheritance with shared Identity By Descent (IBD). This approach contrasts with expectation-maximization algorithms that rely on observed kinship [69] and coalescent analysis-based imputation methods [70], which aim to recreate pedigree relationships. Single Nucleotide Polymorphism (SNP) heritability, reflecting gene content, can be estimated by fitting markers into a mixed model that includes the pedigree as a random effect.

The selection of an appropriate genotype imputation method depends on factors such as trait heritability and the availability of pedigree information. HMM-based methods and RF are preferable for highly heritable traits, while pedigree-based methods like MMM are more suitable for complex traits with low heritability and available pedigree data. Non-parametric methods like KNN and SVD often struggle to accurately represent genotypes due to their inability to capture genetic linkage and inheritance patterns. The impact of imputation methods on the occurrence of repeated markers, which measures linkage disequilibrium, varies across algorithms. HMM-based tools like MaCH and Beagle tend to increase the number of repeated markers as the proportion of missing data increases, whereas non-parametric methods like SVD and KNN do not exhibit this tendency. RF and pedigree-based methods show fewer repeated markers, even with more missing data [68].

B. PHENOTYPE IMPUTATION

In biological studies, phenotypic data are often collected for specific individuals due to cost constraints or difficulties in acquisition. Accurately imputing these phenotypes requires strong genotype-phenotype correlations, as well as the ability to model dependencies across phenotypes. The AutoComplete [71] method has demonstrated exceptional

promise in imputing missing phenotypes, particularly within large biobank datasets. AutoComplete uses an AE architecture to map phenotype vectors into a latent representation and back to the original space, effectively reconstructing the phenotypes. During training, a subset of the original phenotypes is masked to enable the reconstruction of both masked and observed phenotypes with minimal error, employing copy-masking to simulate realistic missingness patterns. AutoComplete outperforms other methods such as KNN, MissForest, MICE, GAIN, HI-VAE, and SoftImpute in phenotype imputation. Moreover, the genotyping positions identified by Genome-Wide Association Studies (GWAS) on imputed phenotypes are consistent with those associated with the originally observed phenotypes, underscoring AutoComplete's robustness and accuracy in preserving true biological signals.

C. SINGLE-CELL RNA DATA IMPUTATION

High-throughput technologies like single-cell RNA-sequencing (scRNA-seq) have revolutionized gene expression analysis at the individual cell level [72]. Unlike bulk RNA sequencing, which aggregates data from multiple cells, scRNA-seq is characterized by high levels of noise and significant sparsity. A major challenge in scRNA-seq data is the presence of false zero values, known as "dropouts", where certain genes go undetected at the single-cell level due to sequencing limitations, capturing only a fraction of the cell's transcripts. These dropouts can lead to the loss of crucial gene-gene relationships, making accurate interpretation of gene expression data difficult.

Imputation techniques designed to recover missing transcripts in single-cell profiles aim to improve the quantification of genes with low or moderate expression levels and to preserve essential gene-gene relationships. Unlike genotype imputation, current scRNA-seq methods rarely use reference maps due to their scarcity, instead drawing information from similar cells or genes within the dataset. For instance, MAGIC [73] employs Markov affinity-based graphs for efficient data diffusion across similar cells, while KNN smoothing [74] aggregates data from neighboring cells based on Poisson statistics. ScImpute [75] calculates the dropout probability for each gene in individual cells and imputes missing values by leveraging data from similar cells, focusing on genes that are less likely to be impacted by dropout events. Despite their effectiveness, these methods can lead to oversmoothing and loss of natural cell-to-cell stochasticity in gene expression. To address this, SAVER [76] employs a Bayesian approach to estimate true expression levels based on gene-to-gene relationships. Additionally, deep learning methods like scIGANs [77], which leverage generative adversarial networks, have been developed to avoid oversmoothing. This technique generates balanced cell representations, enhancing performance for rare cell populations and proving robust for small datasets with few genes exhibiting low expression or cell-to-cell variance.

D. HEALTH RECORD IMPUTATION

Missing data is a significant challenge in health records, primarily due to the diversity of patient information resulting from various types of examinations conducted on different individuals based on their specific diseases. Furthermore, data may be absent due to time constraints during collection or because the patient's condition did not allow for its acquisition. To address the non-random nature of these missing values, it is common practice to add feature values that specify the positions where imputation has been performed [78]. Traditional imputation methods like mean, median, mode, KNN, and MICE are widely used for their simplicity. However, deep learning-based approaches, such as DAEs and GANs, are increasingly preferred for their ability to handle mixed data types and high-dimensional datasets.

E. MEDICAL IMAGE IMPUTATION

Medical imaging faces challenges such as noise and artifacts, which can be mitigated through various denoising techniques, ranging from traditional filtering and statistical methods to advanced deep learning approaches. Low-pass filters [79], [80] effectively reduce noise by replacing sudden peaks with local averages but may compromise the contrast of small lesions. Adaptive filters, such as median [81] and bilateral filters [82], assign weights to neighboring pixels based on statistical characteristics to maintain detail. Numerous enhancements to adaptive filtering have been proposed [83], [84], [85]. Machine learning-based methods [86], [87] and deep learning approaches, particularly convolutional neural networks [88], [89], show great promise in preserving structural information while effectively suppressing noise. In addition to noise and artifacts, medical imaging must also contend with corrupted or missing slices caused by factors like patient motion, aliasing, or signal loss. Traditional interpolation methods attempt to reconstruct missing slices using adjacent ones, but they often lead to blurring and unrealistic results. To address these limitations, GAN-based models have emerged as a powerful solution for medical image imputation.

Super-resolution (SR) aims to reconstruct high-resolution (HR) images from low-resolution (LR) images, restoring texture and structural information. Classical SR methods include interpolation techniques [90] and reconstruction methods [91], [92], while modern approaches leverage neural networks [93]. Recent advances, such as transformer-based models like TTSR [94], highlight the potential of attention mechanisms for superior SR performance.

IV. CONCLUSION

While no single imputation method is suitable for all scenarios, the selection process can be guided by carefully considering factors such as dataset size, the mechanism of missing data, and the specific context. Methods such as MICE, MissForest, MIXGB, and KNN can be impractical for large datasets due to high computational costs. GAIN,

while effective in certain scenarios, often struggles with non-convergence and mode collapse. AEs are particularly useful because they can handle mixed data types and use all incomplete samples during training. Techniques such as missingness augmentation, the inclusion of categorical information, and the application of dot-product attention mechanisms can significantly enhance the quality of imputation.

In genotype imputation, high-quality reference panels with linked haplotypes are essential. The choice of method should consider trait heritability: HMM and RF work well for highly heritable traits, while pedigree-based methods like MMM are better for complex traits with lower heritability and pedigree data. A per-chromosome modeling approach is recommended to create more informative models and to address computational and memory constraints. For phenotype imputation, AutoComplete's deep learning approach excels in accurately imputing missing phenotypes in population-scale biobank datasets.

In single-cell RNA dropout imputation, current scRNA-seq methods typically do not use reference maps due to their limited availability. Instead, they leverage information from similar cells or genes within the observed dataset. Methods like MAGIC, KNN smoothing, ScImpute, and SAVER are effective at recovering missing transcripts and preserving essential gene-gene relationships. To avoid oversmoothing and retain natural cell-to-cell variability in gene expression, generative adversarial networks like scIGANs offer an effective solution.

In health record imputation, using feature indicators to specify imputation positions is recommended, reflecting the non-random nature of missing data in health datasets. For medical imaging, adaptive filters and convolutional neural networks effectively mitigate noise while preserving important artifacts. For slice imputation, GAN-based models excel in generating realistic, anatomically consistent slices, with methods like Pix2Pix and CycleGAN further enhancing performance through label integration and cross-modal learning. Transformer-based models like TTSR demonstrate the power of attention mechanisms for superior SR image reconstruction from LR inputs.

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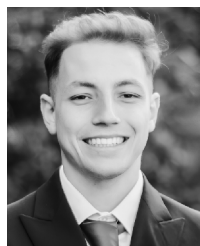
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